

CurioTech Internship

Phase 1 | Month 1 | AI / ML Track

Working Hours: 11:00 AM – 5:00 PM | 6 hrs/day | 5 days/week | ~22 working days

Important: Track all hours on Clockify. Attend Sunday planning and Friday review meetings every week. Learning session with working professionals is held weekly — attendance is mandatory.

Monthly Overview

Week	Theme	Core Topics	Hrs
Week 1	Python & Math Foundations	Python basics, NumPy, Pandas, Linear Algebra, Statistics	30
Week 2	Data Handling & Visualization	Matplotlib, Seaborn, EDA, Feature Engineering, Scikit-learn	30
Week 3	Core ML Algorithms	Regression, Classification, Clustering, Model Evaluation	30
Week 4	Deep Learning & Project	Neural Networks, TensorFlow/Keras, CNNs, Mini-project + GitHub	30

WEEK 1 — Python & Math Foundations

Week Goals: Set up dev environment | Master Python for data science | Understand key math for ML

Day 1 (Sunday) — Environment Setup + Python Refresher [MEETING DAY]

Topic: Dev setup, Python basics, Clockify orientation

Objectives:

- Install Python 3.11, VS Code, Jupyter Notebook, Anaconda
- Set up virtual environment, install numpy/pandas/matplotlib
- Review Python: variables, loops, functions, list comprehensions
- Start Clockify; log 6 hours correctly

Task: Install all tools; write a Python script that prints your name and today's date. Log hours in Clockify.

Resources:

[Python Full Course for Beginners \(freeCodeCamp\)](#)
[Setting Up Python for Data Science \(Corey Schafer\)](#)
[Anaconda Installation Guide](#)

Day 2 (Monday) — NumPy Fundamentals

Topic: NumPy arrays, operations, broadcasting

Objectives:

- Create and manipulate ndarrays
- Understand array indexing, slicing, reshaping
- Perform vectorised mathematical operations
- Broadcasting rules

Task: Complete 15 NumPy exercises: array creation, slicing, math ops, broadcasting.

Resources:

[NumPy Tutorial \(Keith Galli\)](#)
[NumPy Official Quickstart](#)
[NumPy Exercises \(w3resource\)](#)

Day 3 (Tuesday) — Pandas for Data Manipulation

Topic: DataFrames, Series, data wrangling

Objectives:

- Create and load DataFrames (CSV, dict)
- Select, filter, sort, groupby operations
- Handle missing values
- Apply, merge, concat

Task: Load a CSV dataset (e.g. Titanic), clean it, answer 5 data questions using Pandas.

Resources:

[Pandas Tutorial \(Corey Schafer\)](#)
[Pandas Documentation — 10 Minutes](#)
[Pandas Exercises \(Kaggle\)](#)

Day 4 (Wednesday) — Linear Algebra for ML

Topic: Vectors, matrices, dot products, eigenvalues

Objectives:

- Scalars, vectors, matrices — notation and ops
- Matrix multiplication and transpose
- Dot product and its ML significance
- Intro to eigenvalues / eigenvectors

Task: Use NumPy to multiply two matrices, compute dot product, and find eigenvalues of a 3x3 matrix.

Resources:

[Linear Algebra for ML \(3Blue1Brown — Essence of LA\)](#)
[Khan Academy — Linear Algebra](#)
[NumPy Linear Algebra Docs](#)

Day 5 (Thursday) — Statistics & Probability

Topic: Descriptive stats, distributions, probability basics

Objectives:

- Mean, median, mode, variance, standard deviation
- Normal distribution, z-score
- Probability rules, conditional probability, Bayes theorem intro
- Visualise distributions with matplotlib

Task: Use Pandas describe() on a dataset; plot histogram and boxplot; compute z-scores manually.

Resources:

[Statistics for Data Science \(StatQuest\)](#)
[Probability for ML \(StatQuest Playlist\)](#)
[SciPy Stats Tutorial](#)

Friday — Weekly Review Meeting (Check Clockify, discuss Week 1 progress, plan Week 2)

WEEK 2 — Data Handling, Visualisation & ML Intro

Week Goals: Master EDA workflow | Learn Matplotlib & Seaborn | Intro to Scikit-learn

Day 6 (Sunday) — Matplotlib & Seaborn [MEETING DAY]

Topic: Data visualisation fundamentals

Objectives:

- Line, bar, scatter, pie charts with Matplotlib
- Seaborn: heatmap, pairplot, violinplot, countplot
- Customise labels, titles, colours, figure size
- Save figures as PNG/PDF

Task: Create a dashboard of 6 different chart types for the Titanic or Iris dataset.

Resources:

[Matplotlib Tutorial \(Corey Schafer\)](#)
[Seaborn Tutorial \(Kimberly Fessel\)](#)
[Seaborn Official Gallery](#)

Day 7 (Monday) — Exploratory Data Analysis (EDA)

Topic: Systematic EDA pipeline

Objectives:

- Dataset shape, dtypes, null values, duplicates
- Correlation matrix and heatmap
- Outlier detection (IQR, z-score)
- Feature distributions and skewness

Task: Write a complete EDA notebook for the House Prices dataset (Kaggle): shape, nulls, correlations, plots.

Resources:

[EDA in Python — Full Walkthrough \(Rob Mulla\)](#)
[Kaggle EDA Starter Notebook \(Titanic\)](#)
[Pandas Profiling Docs](#)

Day 8 (Tuesday) — Feature Engineering

Topic: Preparing data for ML models

Objectives:

- Encoding categorical variables (Label, One-Hot)
- Feature scaling: StandardScaler, MinMaxScaler
- Handling missing data: imputation strategies
- Creating new features from existing ones

Task: Apply Label Encoding, One-Hot Encoding, and StandardScaler on the Titanic dataset.

Resources:

[Feature Engineering \(Krish Naik\)](#)
[Scikit-learn Preprocessing Docs](#)
[Feature Engineering for ML \(Kaggle Learn\)](#)

Day 9 (Wednesday) — Scikit-learn Introduction

Topic: ML pipeline with Scikit-learn

Objectives:

- Scikit-learn API: fit / predict / score pattern
- Train-test split, cross-validation
- Pipelines: chain preprocessing + model
- Evaluation metrics: accuracy, confusion matrix

Task: Build a Scikit-learn Pipeline: impute → scale → Logistic Regression on Titanic. Report accuracy.

Resources:

[Scikit-learn Tutorial for Beginners \(Sentdex\)](#)
[Scikit-learn Official Getting Started](#)
[Scikit-learn Pipeline Docs](#)

Day 10 (Thursday) — Model Evaluation & Validation

Topic: Measuring model performance correctly

Objectives:

- Accuracy, Precision, Recall, F1-Score
- ROC-AUC curve and interpretation
- Overfitting vs underfitting — bias-variance tradeoff
- K-Fold cross-validation

Task: Evaluate Titanic model: confusion matrix, classification report, ROC-AUC, 5-fold CV score.

Resources:

[Confusion Matrix \(StatQuest\)](#)
[ROC and AUC \(StatQuest\)](#)
[Cross-Validation Explained \(StatQuest\)](#)

Friday — Weekly Review Meeting (Review EDA + Sklearn work, Clockify check, plan Week 3)

WEEK 3 — Core Machine Learning Algorithms

Week Goals: Understand key supervised algorithms | Learn unsupervised clustering | Tune hyperparameters

Day 11 (Sunday) — Linear & Logistic Regression [MEETING DAY]

Topic: Regression and binary classification

Objectives:

- Linear Regression: OLS, gradient descent intuition
- Logistic Regression: sigmoid function, decision boundary
- Regularisation: L1 (Lasso), L2 (Ridge)
- Implement both from scratch with NumPy

Task: Predict house prices with Linear Regression; classify Titanic survivors with Logistic Regression.

Resources:

[Linear Regression \(StatQuest\)](#)
[Logistic Regression \(StatQuest\)](#)
[Regression Tutorial \(Sentdex\)](#)

Day 12 (Monday) — Decision Trees & Random Forest

Topic: Tree-based models

Objectives:

- Decision Tree: Gini impurity, information gain
- Random Forest: bagging, feature importance
- Visualise a decision tree
- Compare single tree vs. ensemble performance

Task: Train Decision Tree and Random Forest on Titanic; plot feature importances; compare accuracies.

Resources:

[Decision Trees \(StatQuest\)](#)
[Random Forest \(StatQuest\)](#)
[Decision Tree Visualisation \(Sklearn\)](#)

Day 13 (Tuesday) — SVM & KNN

Topic: Support Vector Machines and K-Nearest Neighbours

Objectives:

- KNN: distance metrics, choosing K, pros/cons
- SVM: hyperplane, kernel trick (RBF, linear)
- Multi-class classification strategies
- When to use SVM vs KNN

Task: Classify Iris dataset with KNN (test K=3,5,7) and SVM (kernels: linear, rbf). Report best result.

Resources:

[KNN Algorithm Explained \(Simplilearn\)](#)
[SVM Explained \(StatQuest\)](#)
[Sklearn SVM User Guide](#)

Day 14 (Wednesday) — Clustering: K-Means & DBSCAN

Topic: Unsupervised learning fundamentals

Objectives:

- K-Means: centroid algorithm, elbow method
- DBSCAN: density-based, noise points
- Evaluate clustering: silhouette score
- Visualise clusters with PCA reduction

Task: Apply K-Means on Mall Customer Segmentation dataset; use elbow method; visualise with PCA.

Resources:

[K-Means Clustering \(StatQuest\)](#)
[DBSCAN Explained \(Normalized Nerd\)](#)
[Sklearn Clustering Overview](#)

Day 15 (Thursday) — Hyperparameter Tuning & Gradient Boosting

Topic: Optimising models

Objectives:

- GridSearchCV and RandomizedSearchCV
- XGBoost: boosting concept, key hyperparameters
- Early stopping to prevent overfitting
- Compare Random Forest vs XGBoost performance

Task: Use GridSearchCV to tune Random Forest; train XGBoost on Titanic; compare all models in a table.

Resources:

[XGBoost Explained \(StatQuest\)](#)
[GridSearchCV Tutorial \(Krish Naik\)](#)
[XGBoost Official Docs](#)

Friday — Weekly Review Meeting (Demo ML models, Clockify check, plan Week 4 deep learning)

WEEK 4 — Deep Learning, NLP Basics & Mini-Project

Week Goals: Build neural networks with Keras | Understand CNNs | Complete and present mini-project

Day 16 (Sunday) — Neural Networks Fundamentals [MEETING DAY]

Topic: Perceptron → Multi-layer networks

Objectives:

- Biological neuron → artificial neuron
- Activation functions: ReLU, Sigmoid, Softmax
- Forward propagation and loss functions
- Backpropagation and gradient descent

Task: Draw a 3-layer neural net diagram; code forward pass manually in NumPy for XOR problem.

Resources:

[Neural Networks \(3Blue1Brown — Chapter 1\)](#)
[But what IS a Neural Network? \(3Blue1Brown\)](#)
[Backpropagation Explained \(3Blue1Brown\)](#)

Day 17 (Monday) — TensorFlow & Keras

Topic: Building neural networks with Keras

Objectives:

- TensorFlow 2.x and Keras API overview
- Sequential model: Dense layers, compile, fit, evaluate
- Callbacks: ModelCheckpoint, EarlyStopping
- Plot training vs validation loss curves

Task: Build a Keras model to classify MNIST digits; reach >97% test accuracy; plot loss/accuracy curves.

Resources:

[TensorFlow 2.0 Tutorial \(Sentdex\)](#)
[Keras Beginner Tutorial \(Nicholas Renotte\)](#)
[TensorFlow Official Beginner Guide](#)

Day 18 (Tuesday) — Convolutional Neural Networks (CNNs)

Topic: Image classification with CNNs

Objectives:

- Convolution, pooling, feature maps
- CNN architecture: Conv2D → MaxPool → Flatten → Dense
- Data augmentation to prevent overfitting
- Transfer learning concept (intro only)

Task: Build a CNN to classify CIFAR-10 images; add data augmentation; compare with/without augmentation.

Resources:

[CNN Explained \(Deeplizard\)](#)
[CNN for CIFAR-10 \(Nicholas Renotte\)](#)
[Keras CNN Tutorial — TensorFlow Docs](#)

Day 19 (Wednesday) — NLP Basics + Text Classification

Topic: Introduction to Natural Language Processing

Objectives:

- Text preprocessing: tokenisation, stopwords, stemming
- Bag of Words and TF-IDF vectorisation
- Sentiment analysis with Logistic Regression
- Intro to word embeddings concept

Task: Build a movie review sentiment classifier using TF-IDF + Logistic Regression on IMDB dataset.

Resources:

[NLP Tutorial for Beginners \(Krish Naik\)](#)
[Text Classification in Python \(Sentdex\)](#)
[NLTK Official Docs](#)

Day 20 (Thursday) — Mini-Project Day 1: Planning & Data

Topic: Start end-to-end ML project

Objectives:

- Choose project: Spam Detection, Disease Prediction, or House Price Prediction
- Collect/download dataset; perform full EDA
- Feature engineering and preprocessing pipeline
- Baseline model training

Task: Commit project folder to GitHub: dataset, EDA notebook, preprocessing script, baseline model.

Resources:

[End-to-End ML Project \(Krish Naik\)](#)
[Kaggle Datasets](#)
[UCI ML Repository](#)

Friday — Weekly Review Meeting (Mini-project check-in; discuss blockers; confirm Day 21 deliverables)

Day 21 (Saturday Optional / Make-up Day) — Mini-Project Day 2: Build, Evaluate, Document

Topic: Complete and present mini-project

Objectives:

- Train and tune best model (GridSearch / XGBoost / Keras)
- Evaluate with full metrics: accuracy, F1, ROC-AUC
- Write project README with problem, approach, results
- Push final code to GitHub; record a short demo

Task: Submit: GitHub repo link with README, notebook, model file, and results table to the team.

Resources:

[How to Write a Good README \(freeCodeCamp\)](#)
[GitHub for Beginners \(Traversy Media\)](#)
[ML Project Structure Best Practices \(Towards Data Science\)](#)

Essential Resource Library

All links below are stable, publicly available resources. Bookmark them all on Day 1.

Category	Name	URL
Video	Python Full Course (freeCodeCamp)	youtube.com/watch?v=rfscVS0vtbw
Video	NumPy Tutorial (Keith Galli)	youtube.com/watch?v=QUT1VHiLmml
Video	Pandas Tutorial (Corey Schafer)	youtube.com/watch?v=ZyhVh-qRZPA
Video	Matplotlib (Corey Schafer)	youtube.com/watch?v=UO98IJQ3QGI
Video	StatQuest ML Series (Josh Starmer)	youtube.com/@statquest
Video	3Blue1Brown — Neural Networks	youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
Video	3Blue1Brown — Linear Algebra	youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFItgF8hE_ab
Video	Krish Naik — ML/DL Playlist	youtube.com/@krishnaik06
Video	Sentdex — Python for Finance/ML	youtube.com/@sentdex
Video	Deeplizard — Keras & PyTorch	youtube.com/@deeplizard
Docs	NumPy Official Docs	numpy.org/doc/stable/
Docs	Pandas Official Docs	pandas.pydata.org/docs/
Docs	Scikit-learn User Guide	scikit-learn.org/stable/user_guide.html
Docs	TensorFlow Tutorials	tensorflow.org/tutorials
Docs	Keras API Reference	keras.io/api/
Course	Kaggle Learn (free micro-courses)	kaggle.com/learn
Course	Google ML Crash Course	developers.google.com/machine-learning/crash-course
Book	Hands-On ML (free preview)	github.com/ageron/handson-ml3
Data	Kaggle Datasets	kaggle.com/datasets
Data	UCI ML Repository	archive.ics.uci.edu/
Tool	Clockify Time Tracker	clockify.me
Tool	GitHub	github.com
Tool	Google Colab (free GPU)	colab.research.google.com

Clockify Rules Reminder — Non-compliance = hours not counted Start timer at 11:00 AM sharp. Stop at 5:00 PM. Tag correct project. No fake hours. Missing records = absent. Attend Sunday + Friday meetings every week without exception.